182 Mark for Review

Which expression is equivalent to $5xy + 30x^2y + 35xy^2$?

Ⓐ $5xy(1 + 30x + 35y)$

Ⓑ $5xy(1 + 6x + 7y)$

Ⓒ $5xy(30x + 35y)$

Ⓓ $5xy(6x + 7y)$

Answer: B

183 Mark for Review

A distance of 86 furlongs is equivalent to how many feet?
$(1 \text{ furlong} = 220 \text{ yards and } 1 \text{ yard} = 3 \text{ feet})$

56760

Answer: (grid-in: 56760)

184 Mark for Review

The product of a number and $-\frac{7}{15}$ is 1. What is the number?

Ⓐ $\frac{15}{7}$

Ⓑ $\frac{7}{15}$

Ⓒ $-\frac{7}{15}$

Ⓓ $-\frac{15}{7}$

Answer: D

185 Mark for Review

A plant nursery classifies a perennial flower species as tall if its typical height when fully grown is more than 100 centimeters. Each of the following inequalities represents the possible heights h, in centimeters, for a specific perennial flower species when fully grown. Which inequality represents the possible heights h, in centimeters, for a tall perennial flower species?

(A) $18 < h < 94$

(B) $42 < h < 87$

(C) $75 < h < 100$

(D) $110 < h < 151$

Answer: D

186 Mark for Review

Which expression is equivalent to $(7x^4 + x^2) + (x^4 + 8x^2)$?

(A) $8x^8 + 9x^4$

(B) $7x^{16} + 8x^4$

(C) $8x^4 + 9x^2$

(D) $8x^8 + 7x^6$

Answer: C

187 Mark for Review

The equation $E(t) = 6(1.7)^t$ gives the estimated number of employees at a certain business, where t is the number of years since the business opened. Which of the following is the best interpretation of the number 6 in this context?

Ⓐ The increase in the estimated number of employees each year

Ⓑ The estimated number of employees when the business opened

Ⓒ The percent increase in the estimated number of employees each year

Ⓓ The number of years the business has been open

Answer: B

188 Mark for Review

Each angle of $\triangle ABC$ is congruent to one of the angles of $\triangle QRS$. If $AB = 42$, $BC = 63$, $AC = 84$, and $RS = 252$, what is one possible value for the perimeter of $\triangle QRS$?

567

Answer: (grid-in: 567)

189 Mark for Review

$$17x + 16y = c$$
$$16x - 17y = c$$

In the given system of equations, c is a positive constant. Which of the following could be the point where the graphs of the equations in this system intersect in the xy-plane?

A. $\left(10, -\frac{10}{33}\right)$

B. $(10, 330)$

C. $\left(c, \frac{c}{33}\right)$

D. $(c, 0)$

Answer: A

190 Mark for Review

$$2|x-2|=14$$

What is the sum of the solutions to the given equation?

A) -7

B) 0

C) 4

D) 16

Answer: C

191 Mark for Review

A right square prism has a height of 9 units. The volume of the prism is 324 cubic units. What is the length, in units, of an edge of the base?

A) 6

B) 9

C) 18

D) 36

Answer: A

192 Mark for Review

If $4+\frac{5(11-x)}{7}=3(11-x)+4$, what is the value of $11+x$?

22

Answer: (grid-in: 22)

193 Mark for Review

A book publishing company pays the author of a certain book $\$3.50$ per book for the first 400 books sold. After the first 400 books are sold, the payment increases to $\$4.75$ per book sold. Which function gives the author's total payment $P(b)$, in dollars, in terms of the number of books sold, b, where $b > 400$?

A) $P(b) = 3.50(400) + 4.75b$

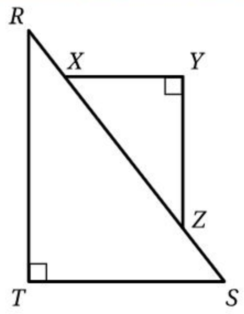
B) $P(b) = 3.50b + 4.75b$

C) $P(b) = 3.50b + 4.75(b - 400)$

D) $P(b) = 3.50(400) + 4.75(b - 400)$

Answer: D

194 Mark for Review



Note: Figure not drawn to scale.

In triangles RST and XYZ shown, $\overline{XY}$ is parallel to $\overline{TS}$ and $\tan R = \frac{252}{275}$. What is the value of $\sin X$ in triangle XYZ?

Ⓐ $\frac{252}{527}$

Ⓑ $\frac{252}{373}$

Ⓒ $\frac{275}{373}$

Ⓓ $\frac{275}{252}$

Answer: C

195 Mark for Review

$$x + 5 = 2(5)$$

What value of x is the solution to the given equation?

(A) -10

(B) -5

(C) 5

(D) 10

Answer: C

196 Mark for Review

What is 20% of 240?

Ⓐ 48

Ⓑ 96

Ⓒ 200

Ⓓ 220

Answer: A

197 Mark for Review

$$3x = 9 + 8y$$
$$-3x = 2 + 8y$$

The solution to the given system of equations is (x, y). What is the value of $6x$?

7

Answer: (grid-in: 7)

198 Mark for Review

$$y \geq 25$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

(A)

x	y
0	-25
1	-25
2	-25

(B)

x	y
0	0
1	0
2	0

(C)

x	y
0	24
1	24

Answer: D

199 Mark for Review

A triangle has a base length of 20 centimeters and a corresponding height of 80 centimeters. What is the area, in square centimeters, of the triangle?

Ⓐ 100

Ⓑ 400

Ⓒ 800

Ⓓ $1,600$

Answer: C

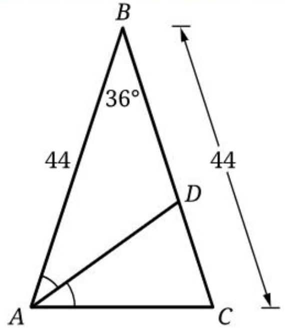
200 Mark for Review

A survey of the daily commuting times, in minutes, of seven different Chicago-based employees at a certain company yielded a data set consisting of the values 32, 14, 43, 33, 51, 24, and 45. A survey of the daily commuting times, in minutes, of seven different San Francisco–based employees at the same company yielded a data set consisting of the values 47, 32, 17, 34, 47, x, and 37. If the means of the two data sets are equal, what is the value of x?

28

Answer: (grid-in: 28)

201 Mark for Review



Note: Figure not drawn to scale.

For isosceles triangle ABC, $AB = BC = 44$, the measure of angle ABC is 36°, point D lies on $\overline{BC}$, and $\angle BAC$ is bisected by $\overline{AD}$. Which of the following statements must be true?

A) $AB = AC = BC$

B) $AD = BD = AC$

C) $BD = CD = AC$

D) $AB = BD = AD$

Answer: B

202 Mark for Review

What is the solution to the equation $3\sqrt{x-15}=\sqrt{5}$?

(A) $\frac{140}{9}$

(B) $\frac{160}{9}$

(C) $\frac{20}{3}$

(D) $\frac{50}{3}$

Answer: A

203 Mark for Review

$$n(x) = 2(x + 7) + (x - 7)$$

The function n is defined as shown. For what value of x is $n(x) = 67$?

20

Answer: (grid-in: 20)

204 Mark for Review

A study established that in a certain park, there are at least 141 squirrels, consisting of red squirrels and gray squirrels. The study also established that there are at least 4 times as many gray squirrels as red squirrels in this park. Which of the following systems of inequalities best represents this situation, where g is the number of gray squirrels in this park and r is the number of red squirrels in this park?

A) $g + r \leq 141$
$g \geq 4r$

B) $g + r \leq 141$
$g \leq 4r$

C) $g + r \geq 141$
$g \geq 4r$

D) $g + r \geq 141$
$g \leq 4r$

Answer: C

205 Mark for Review

A company has a customer loyalty program. In January 2018, there were $1{,}100$ customers enrolled in the loyalty program. For the next 24 months after January 2018, the total number of customers enrolled in the loyalty program each month was 4% greater than the total number enrolled the previous month. Which equation gives the total number of customers, c, enrolled in the company's loyalty program m months after January 2018, where $m \leq 24$?

(A) $c = 1{,}100(0.04)^m$

(B) $c = 1{,}100(1.04)^m$

(C) $c = 1{,}100(1.4)^m$

(D) $c = 1{,}100(4)^m$

Answer: B

206 Mark for Review

The measure of angle F is $\frac{\pi}{12}$ radians. If the measure of angle F is $3n$ **degrees**, where n is a constant, what is the value of n?

Ⓐ 3

Ⓑ 5

Ⓒ 12

Ⓓ 15

Answer: B

207 Mark for Review

$$5x^2 + 2x - 6 = 0$$

What is one of the solutions to the given equation?

A) $\frac{1-\sqrt{31}}{10}$

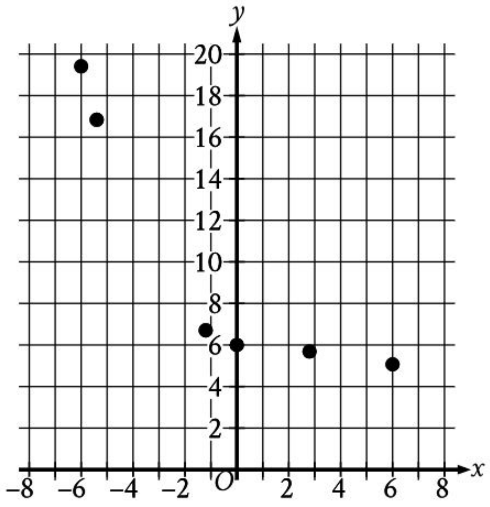
B) $\frac{1-\sqrt{31}}{5}$

C) $\frac{-1-\sqrt{31}}{10}$

D) $\frac{-1-\sqrt{31}}{5}$

Answer: D

208 Mark for Review



Which of the following equations is the most appropriate model for the data shown in the scatterplot?

Ⓐ $y = -(1.56)^{-x} + 6$

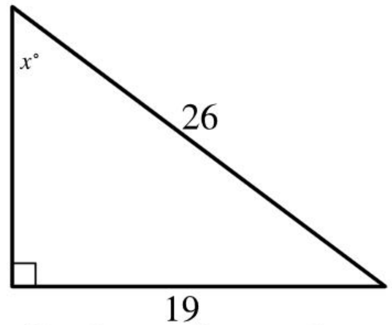
Ⓑ $y = -(1.56)^{x} + 6$

Ⓒ $y = (1.56)^{-x} - 6$

Ⓓ $y = (1.56)^{-x} + 5$

Answer: D

209 Mark for Review



Note: Figure not drawn to scale.

In the triangle shown, what is the value of $\sin x^\circ$?

0.730769230769

Answer: —

210 Mark for Review

$$y = 4x^2 - 24x + 42$$
$$y + 3 = 0$$

How many solutions are there to the given system of equations?

Ⓐ There is exactly 1 solution.

Ⓑ There are exactly 2 solutions.

Ⓒ There are exactly 3 solutions.

Ⓓ There are no solutions.

Answer: D

211 Mark for Review

A jar has 770 marbles, and 20% of these marbles are orange. How many marbles in the jar are orange?

(A) 154

(B) 308

(C) 385

(D) 750

Answer: A

212 Mark for Review

$$y = 24$$
$$y - x = 0$$

Which of the following points (x, y) represents the solution to the given system of equations in the *xy*-plane?

Ⓐ $(0, 0)$

Ⓑ $(0, 24)$

Ⓒ $(24, 0)$

Ⓓ $(24, 24)$

Answer: D

213 Mark for Review

The function g is defined by $g(x) = |10x| + 15$. What is the value of $g(-6)$?

75

Answer: (grid-in: 75)

214 Mark for Review

For the linear function g, $g(4) = 10$ and $g(x + 1) - g(x) = 6$. Which equation defines g?

(A) $g(x) = 4x + 10$

(B) $g(x) = 4x + 6$

(C) $g(x) = 6x - 14$

(D) $g(x) = 6x - 24$

Answer: C

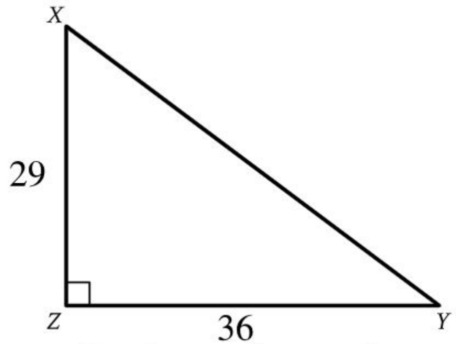
215 Mark for Review

The sides of a rectangle are such that the length is 3 centimeters more than 5 times the width, in centimeters. If the width of the rectangle is 11 centimeters, what is the area, in square centimeters, of the rectangle?

638

Answer: (grid-in: 638)

216 Mark for Review



Note: Figure not drawn to scale.

Triangle XYZ shown is a right triangle. Which of the following has the same value as $\sin X$?

A) $\tan X$

B) $\tan Y$

C) $\cos X$

D) $\cos Y$

Answer: D

217 Mark for Review

$$f(x) = (2x - 13)(2x + 5)$$

What is the minimum value of the given function?

(A) -81

(B) $-\frac{5}{2}$

(C) 2

(D) 8

Answer: A

218 Mark for Review

A rectangle has a perimeter of 42 inches. If the sum of the lengths of three sides of the rectangle is 30 inches, what is the length, in inches, of the fourth side?

(A) 9

(B) 12

(C) 42

(D) 72

Answer: B

219 Mark for Review

Casey is raising h horses and g goats. Each horse requires 2 acres of land, and each goat requires 1 acre of land. The h horses and g goats require a total of 79 acres of land. Which equation represents this situation?

A. $2g + h = 79$

B. $g + 2h = 79$

C. $g + \frac{h}{2} = 79$

D. $\frac{g}{2} + h = 79$

Answer: B

220 Mark for Review

$$x + 80 = 90$$

What value of x is the solution to the given equation?

A) -170

B) -10

C) 10

D) 170

Answer: C

221 Mark for Review

What length, in meters, is equivalent to a length of 210 decimeters?
(1 meter $=$ 10 decimeters)

Ⓐ 2.1

Ⓑ 21

Ⓒ 210

Ⓓ 2,100

Answer: B

222 Mark for Review

$$171, 182, 255, 275, 290$$

What is the minimum value of the data set shown?

A) 119

B) 171

C) 255

D) 290

Answer: B

223 Mark for Review

The function j is defined by $j(x) = \frac{x}{8}$. What is the value of $j(40)$?

(A) 40

(B) 32

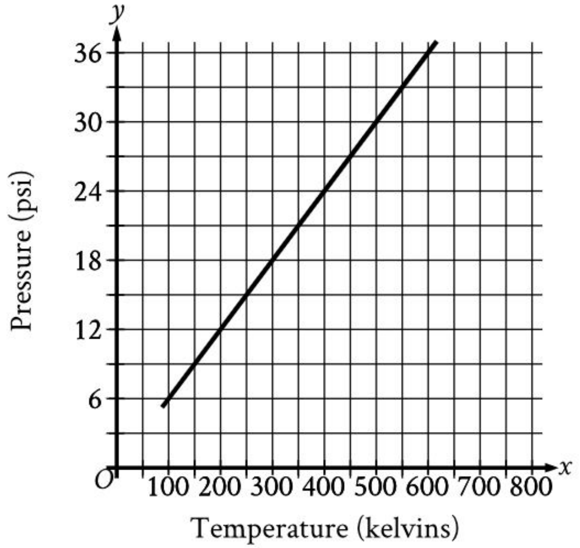
(C) 8

(D) 5

Answer: D

224 Mark for Review

A certain gas is placed inside a container with a constant volume. The graph shows the estimated pressure y, in pounds per square inch (psi), of the gas when its temperature is x kelvins.



What is the estimated pressure of the gas, in psi, when the temperature is 500 kelvins?

(A) 15

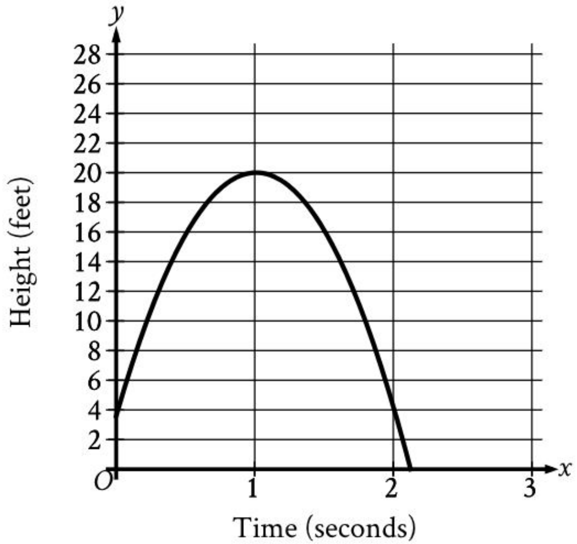
(B) 30

(C) 250

(D) 500

Answer: B

225 Mark for Review



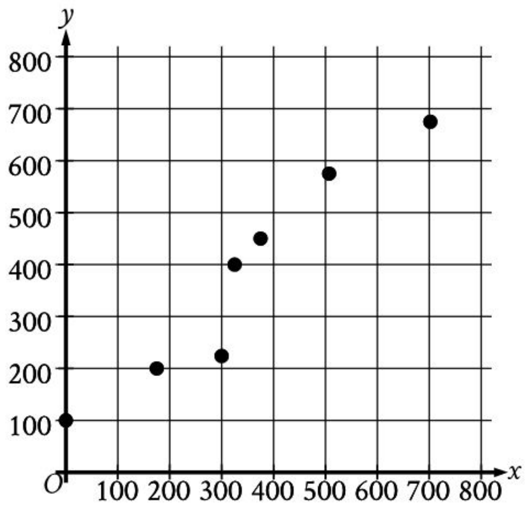
During a physics demonstration, an object was thrown upward and then fell to the ground. The graph shows the height above the ground y, in feet, of the object x seconds after it was thrown. To the nearest whole number, what was the total distance, in feet, that the object fell after reaching its highest point above the ground?

20

Answer: (grid-in: 20)

226 Mark for Review

The scatterplot shows the relationship between two variables, x and y.



Which of the following equations is the most appropriate linear model for the data shown?

Ⓐ $y = 70.20 + 0.89x$

Ⓑ $y = 0.89 + 70.20x$

Ⓒ $y = 70.20x$

Ⓓ $y = -0.89x$

Answer: A

227 Mark for Review

If x is a positive integer, the expression $0.42x$ represents what percent of x?

Ⓐ 42

Ⓑ 58

Ⓒ 142

Ⓓ 158

Answer: A

228 Mark for Review

Natalie bought both neon tetra fish and swordtail fish for a total of $\$34$. Each neon tetra fish costs $\$5$ and each swordtail fish costs $\$7$. If Natalie bought 2 swordtail fish, how many neon tetra fish did she buy?

(A) 4

(B) 12

(C) 14

(D) 32

Answer: A

229 Mark for Review

Each rock in a collection of 50 rocks was classified as either igneous, metamorphic, or sedimentary, as shown in the frequency table.

Classification	Frequency
igneous	10
metamorphic	28
sedimentary	12

If one of these rocks is selected at random, what is the probability of selecting a rock that is igneous?

A) $\frac{10}{12}$

B) $\frac{10}{28}$

C) $\frac{10}{40}$

D) $\frac{10}{50}$

Answer: D

230 Mark for Review

Which expression is equivalent to $4x^3 + 32w$?

(A) $4(x^3 + 8w)$

(B) $4(x^3 + 32w)$

(C) $8(x^3 + 4w)$

(D) $32(x^3 + w)$

Answer: A

231 Mark for Review

An event planner is planning an ice-skating event. It costs the event planner a onetime fee of $\$35$ to rent the skating rink and $\$12.75$ per attendee. The event planner has a budget of $\$200$. What is the greatest number of attendees possible without exceeding the budget?

12

Answer: (grid-in: 12)

232 Mark for Review

Points F and G lie on a circle with center H. Segment FG is a diameter of the circle. If the length of segment FG is 46 centimeters, what is the length, in centimeters, of segment FH?

23

Answer: (grid-in: 23)

233 Mark for Review

$$f(x) = x^3 + 7x + 18$$

For the given function f, the graph of $y = f(x)$ in the xy-plane passes through the point $(0, b)$, where b is a constant. What is the value of b?

18

Answer: (grid-in: 18)

234 Mark for Review

$$2x = 7 + 8y$$
$$-2x = 4 + 8y$$

The solution to the given system of equations is (x, y). What is the value of $4x$?

3

Answer: (grid-in: 3)

235 Mark for Review

$$f(x) = 3x + b$$

For the linear function f, b is a constant and $f(4) = 12$. What is the value of b?

(A) 0

(B) 1

(C) 3

(D) 4

Answer: A

236 Mark for Review

The graph of $y = \left(\frac{1}{7}\right)^x + 1$ in the *xy*-plane is shifted up 5 units. What is an equation of the resulting graph?

Ⓐ $y = \left(\frac{1}{7}\right)^{(x+5)} + 1$

Ⓑ $y = \left(\frac{1}{7}\right)^{(x-5)} + 1$

Ⓒ $y = 5\left(\frac{1}{7}\right)^x + 1$

Ⓓ $y = \left(\frac{1}{7}\right)^x + 6$

Answer: D

237 Mark for Review

The function $P(x) = 517(1.011)^x$ models the population of a certain town each year from 1956 through 1971, where x is the number of years after 1956. Which of the following is the best interpretation of "$P(5)$ is approximately equal to 546" in this context?

(A) The population of the town is estimated to be approximately 546 in 1961.

(B) The population of the town is estimated to be approximately 5 greater in 1961 than in 1956.

(C) The population of the town is estimated to be approximately 5 times greater in 1961 than in 1956.

(D) The population of the town is estimated to increase by approximately 546 every 5 years between 1956 and 1971.

Answer: A

238 Mark for Review

$$x^2 - 225 = 0$$

How many distinct real solutions does the given equation have?

(A) Zero

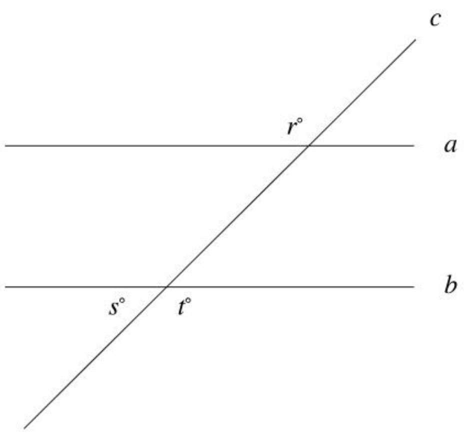
(B) Exactly one

(C) Exactly two

(D) Infinitely many

Answer: C

239 Mark for Review



Note: Figure not drawn to scale.

In the figure, lines a and b are parallel and line c intersects both lines. If $r = 8k - 20$ and $s = 6k - 38$, what is the value of t?

116

Answer: (grid-in: 116)

240 Mark for Review

What length, in meters, is equivalent to a length of 170 decimeters?
$(1 \text{ meter} = 10 \text{ decimeters})$

17

Answer: (grid-in: 17)

241 Mark for Review

168, 191, 259, 262, 286

What is the minimum value of the data set shown?

(A) 118

(B) 168

(C) 259

(D) 286

Answer: B

242 Mark for Review

The function j is defined by $j(x) = \frac{x}{9}$. What is the value of $j(45)$?

(A) 45

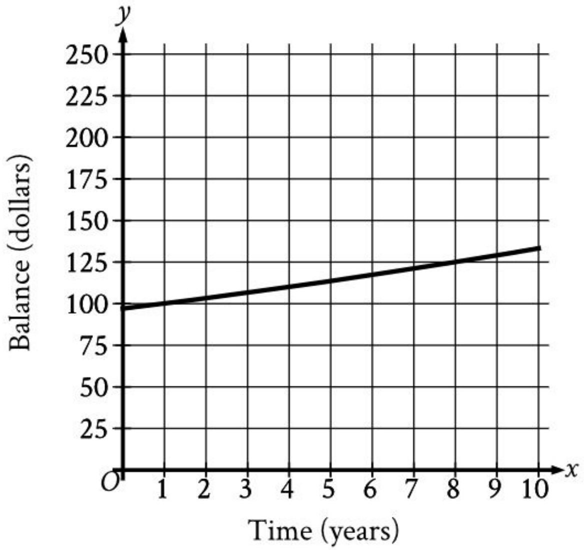
(B) 36

(C) 9

(D) 5

Answer: D

243 Mark for Review



The graph shows the estimated balance of a bank account, in dollars, x years after an initial deposit is made. According to the graph, which of the following is closest to the number of years it took for the estimated balance to increase from the initial deposit to 125 dollars?

A) 0

B) 8

C) 100

D) 125

Answer: B

244 Mark for Review

A botanist studied the growth rate of a birch tree over two years. In the first year, the height of the tree increased c inches. In the second year, the height of the tree increased d inches. The height of the tree increased a total of 53 inches over these two years. Which equation represents this situation?

(A) $c + d = 53$

(B) $c - d = 53$

(C) $-c + d = 53$

(D) $-c - d = 53$

Answer: A

245 Mark for Review

$$\frac{x}{6} = 5$$

What value of x is the solution to the given equation?

(A) 6

(B) 11

(C) 30

(D) 60

Answer: C

246 Mark for Review

Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. If $a = 29$, what is the value of b?

Ⓐ 389

Ⓑ 151

Ⓒ 29

Ⓓ -61

Answer: C

247 Mark for Review

For a psychology class, a student's research paper needs to have no more than 5,000 words. If a student has already written 3,022 words, what is the maximum number of additional words the student can write for the research paper?

A) 1,978

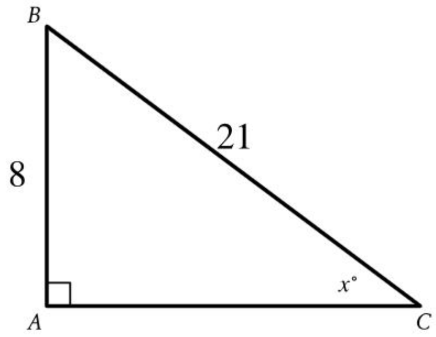
B) 3,022

C) 5,000

D) 8,022

Answer: A

248 Mark for Review



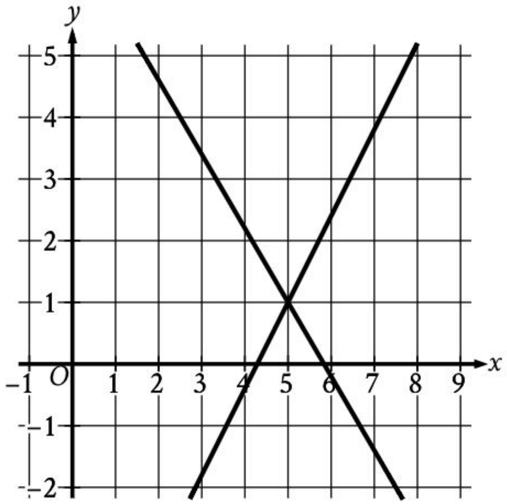
Note: Figure not drawn to scale.

The expression $k \cdot \cos x^\circ$ represents the area of triangle ABC shown. What is the value of k?

84

Answer: (grid-in: 84)

249 Mark for Review



The graph of a system of linear equations is shown. The solution to the system is (x, y). What is the value of x?

5

Answer: (grid-in: 5)

250 Mark for Review

$$y \geq x + 15$$
$$y \leq -5x - 15$$

Which point (x, y) is a solution to the given system of inequalities in the xy-plane?

(A) $(-4, 8)$

(B) $(-4, 11)$

(C) $(-6, 8)$

(D) $(-6, 11)$

Answer: D

251 Mark for Review

$$x = 6$$

$$x + 7y = 27$$

The solution to the given system of equations is (x, y). What is the value of y?

3

Answer: (grid-in: 3)

252 Mark for Review

A model estimates that the initial number of photons in an X-ray beam is 500 when the beam reaches the surface of a certain material. The model also estimates that when the beam passes through this material, the number of photons in the beam decreases by 50% for each 8 millimeters of the material the beam passes through. Which equation represents this model, where I is the estimated number of photons in the beam when the beam reaches a point x millimeters from the surface?

A) $I = 500(0.5)^{\frac{x}{8}}$

B) $I = 500(0.5)^{x}$

C) $I = 500(2)^{\frac{x}{8}}$

D) $I = 500(8)^{\frac{x}{2}}$

Answer: A

253 Mark for Review

x	y
0	80
1	90
2	100

For a linear relationship between x and y, the table gives three values of x and their corresponding values of y. Which equation represents this relationship?

A) $y = 10x + 80$

B) $y = x + 90$

C) $y = x + 10$

D) $y = 10x$

Answer: A

254 Mark for Review

At a convention center, there are a total of 325 visitors. Each visitor is located in either room A, room B, or room C. If one of these visitors is selected at random, the probability of selecting a visitor who is located in room A is 0.68, and the probability of selecting a visitor who is located in room B is 0.28. How many visitors are located in room C?

(A) 4

(B) 13

(C) 62

(D) 130

Answer: B

255 Mark for Review

$$5x = 15yz + 10$$

The given equation relates the distinct, positive real numbers x, y, and z. Which equation correctly expresses z in terms of x and y?

A) $z = \frac{5x}{10} - 15y$

B) $z = \frac{5x}{15y} - 10$

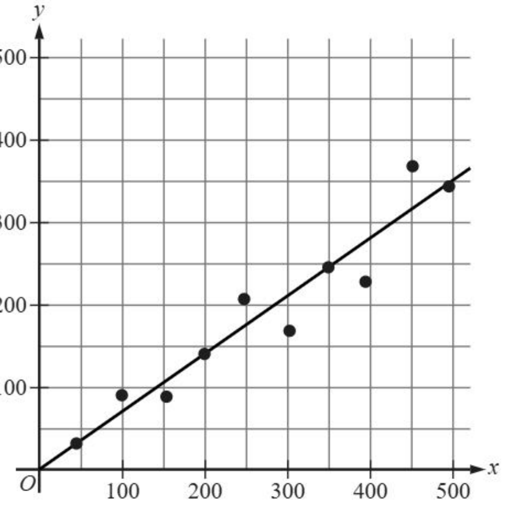
C) $z = \frac{5x-10}{15y}$

D) $z = \frac{5x-15y}{10}$

Answer: C

256 Mark for Review

A group of 10 gardeners recorded data on the germination rates of their tomato crop for one growing season. The scatterplot shows the relationship between the number of tomato seeds planted, x, and the number of tomato seeds that germinated, y, for each of the gardeners. A line of best fit is also shown.



Which of the following is the best interpretation of the slope of the line of best fit in this context?

(A) The number of tomato seeds planted is predicted to increase by 70 seeds every 100 days.

(B) The number of tomato seeds planted is predicted to increase by 350 seeds every 100 days.

(C) The number of tomato seeds that germinate is predicted to increase by 70 seeds for every additional 100 tomato seeds that are planted.

Answer: C

257 Mark for Review

A circle in the *xy*-plane has its center at $(2, 1)$ and has a radius of length 9. Which equation represents this circle?

A. $(x+2)^2+(y+1)^2=9$

B. $(x-2)^2+(y-1)^2=9$

C. $(x+2)^2+(y+1)^2=81$

D. $(x-2)^2+(y-1)^2=81$

Answer: D

258 Mark for Review

A rectangle has a perimeter of 42 inches. If the sum of the lengths of three sides of the rectangle is 26 inches, what is the length, in inches, of the fourth side?

A) 5

B) 16

C) 42

D) 68

Answer: B

259 Mark for Review

Casey is raising h horses and g goats. Each horse requires 2 acres of land, and each goat requires 1 acre of land. The h horses and g goats require a total of 46 acres of land. Which equation represents this situation?

(A) $2g + h = 46$

(B) $g + 2h = 46$

(C) $g + \frac{h}{2} = 46$

(D) $\frac{g}{2} + h = 46$

Answer: B

260 Mark for Review

$$x + 90 = 110$$

What value of x is the solution to the given equation?

(A) -200

(B) -20

(C) 20

(D) 200

Answer: C

261 Mark for Review

$$73, 76, 78, 79, 81, 84, 85, 86, 96$$

What is the median of the data shown?

81

Answer: (grid-in: 81)

262 Mark for Review

What length, in centimeters, is equivalent to a length of 75 meters?
$(1 \text{ meter} = 100 \text{ centimeters})$

(A) 0.075

(B) 0.75

(C) 7,500

(D) 75,000

Answer: C

263 Mark for Review

Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. If $a = 10$, what is the value of b?

A) -80

B) 10

C) 170

D) 370

Answer: B

264 Mark for Review

If x is a positive integer, the expression $0.39x$ represents what percent of x?

(A) 161

(B) 139

(C) 61

(D) 39

Answer: D

265 Mark for Review

Which expression is equivalent to $2x^5 + 12w$?

Ⓐ $2(x^5 + 6w)$

Ⓑ $2(x^5 + 12w)$

Ⓒ $6(x^5 + 2w)$

Ⓓ $12(x^5 + w)$

Answer: A

266 Mark for Review

$$x^2 + 5 = (17)^2 + 5$$

What is a solution to the given equation?

17

Answer: (grid-in: 17)

267 Mark for Review

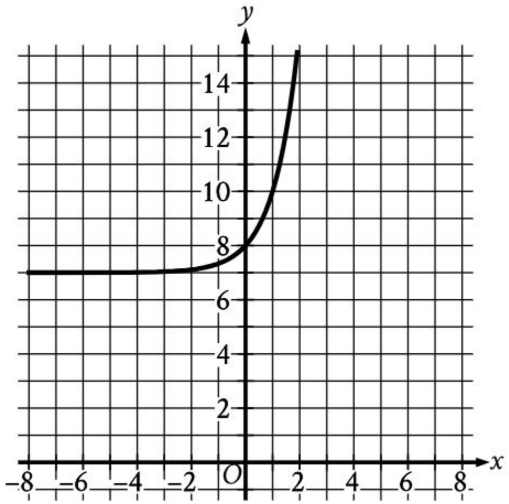
$$f(x) = 35x + 500$$

The function f gives the monthly fee $f(x)$, in dollars, a storage company charges to keep x boxes in its warehouse. What is the monthly fee, in dollars, the company charges to keep 60 boxes in its warehouse?

2600

Answer: (grid-in: 2600)

268 Mark for Review



What is an equation of the graph shown?

Ⓐ $y = 3^x + 7$

Ⓑ $y = 3^{-x} + 7$

Ⓒ $y = 3^x + 8$

Ⓓ $y = 3^{-x} + 8$

Answer: A

269 Mark for Review

If the equations $-8x + y = 19$ and $8x + y = 19$ are graphed in the *xy*-plane, at what point (x, y) do the graphs intersect?

Ⓐ $(0, 19)$

Ⓑ $(0, 8)$

Ⓒ $(8, 0)$

Ⓓ $(19, 0)$

Answer: A

270 Mark for Review

A rectangle has a length that is 31 times its width. The function $y = (31w)(w)$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. Which of the following is the best interpretation of $31w$ in this context?

A. The length of the rectangle, in feet

B. The area of the rectangle, in square feet

C. The difference between the length and the width of the rectangle, in feet

D. The width of the rectangle, in feet

Answer: A

271 Mark for Review

$$x + 14 = y(z + 8)$$

The given equation relates the distinct positive variables x, y, and z. Which equation correctly represents z in terms of x and y?

(A) $z = \frac{x+6}{y}$

(B) $z = \frac{x}{y} + 6$

(C) $z = \frac{x+14}{y} + 8$

(D) $z = \frac{x+14}{y} - 8$

Answer: D

272 Mark for Review

$$y = \frac{1}{8}x - 22$$
$$y = -x + 32$$

The solution to the given system of equations is (x, y). What is the value of x?

48

Answer: (grid-in: 48)

273 Mark for Review

The graph of radical function f in the xy-plane, where $y = f(x)$, contains the points $(-5, 11)$, $(-2.6, 0)$, and $(0, -9.5)$. What is the value of $f(0)$?

-9.5

Answer: (grid-in: -9.5)

274 Mark for Review

A rectangle has a perimeter of 46 inches. If the sum of the lengths of three sides of the rectangle is 32 inches, what is the length, in inches, of the fourth side?

(A) 9

(B) 14

(C) 46

(D) 78

Answer: B

275 Mark for Review

The height of a right circular cylinder is 21 inches, and the circumference of its base is 210 inches. Which expression represents the total surface area, in square inches, of the cylinder?

Ⓐ $(21)(210) + 2\pi\left(\frac{105}{\pi}\right)^2$

Ⓑ $(21)(210) + \pi\left(\frac{105}{\pi}\right)^2$

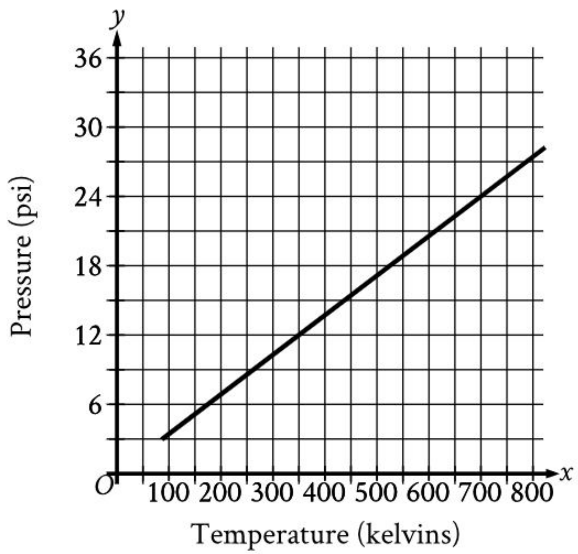
Ⓒ $\pi\left(\frac{105}{\pi}\right)^2(21)$

Ⓓ $(21)(210)$

Answer: A

276 Mark for Review

A certain gas is placed inside a container with a constant volume. The graph shows the estimated pressure y, in pounds per square inch (psi), of the gas when its temperature is x kelvins.



What is the estimated pressure of the gas, in psi, when the temperature is 700 kelvins?

Ⓐ 700

Ⓑ 350

Ⓒ 24

Ⓓ 12

Answer: C

277 Mark for Review

A botanist studied the growth rate of a birch tree over two years. In the first year, the height of the tree increased c inches. In the second year, the height of the tree increased d inches. The height of the tree increased a total of 65 inches over these two years. Which equation represents this situation?

(A) $-c-d=65$

(B) $-c+d=65$

(C) $c-d=65$

(D) $c+d=65$

Answer: D

278 Mark for Review

Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. If $a = 24$, what is the value of b?

(A) -66

(B) 24

(C) 156

(D) 384

Answer: B

279 Mark for Review

If x is a positive integer, the expression $0.15x$ represents what percent of x?

(A) 185

(B) 115

(C) 85

(D) 15

Answer: D

280 Mark for Review

Jonathon bought both neon tetra fish and swordtail fish for a total of $51. Each neon tetra fish costs $5 and each swordtail fish costs $7. If Jonathon bought 3 swordtail fish, how many neon tetra fish did he buy?

(A) 6

(B) 12

(C) 21

(D) 48

Answer: A

281 Mark for Review

$$f(x) = x + \frac{4}{9}$$

The function f is defined by the given equation. What is the value of $f(x)$ when $x = \frac{5}{9}$?

A. $\frac{4}{9}$

B. $\frac{5}{9}$

C. 1

D. 9

Answer: C

282 Mark for Review

A car dealership has only sedans, SUVs, and minivans for sale. On Monday, 20% of the vehicles for sale were sedans and 50% were SUVs. If there were 36 sedans for sale at the dealership on Monday, how many minivans were for sale?

54

Answer: (grid-in: 54)

283 Mark for Review

$$y \geq 12$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A

x	y
0	-12
1	-12
2	-12

B

x	y
0	0
1	0
2	0

C

x	y
0	11
1	11

Answer: D

284 Mark for Review

$$x + 4 = 12$$
$$(x + 4)^2 = y$$

Which ordered pair (x, y) is a solution to the given system of equations?

(A) $(8, 144)$

(B) $(8, 8)$

(C) $(8, 12)$

(D) $(8, 48)$

Answer: A

285 Mark for Review

At a convention center, there are a total of 225 visitors. Each visitor is located in either room A, room B, or room C. If one of these visitors is selected at random, the probability of selecting a visitor who is located in room A is 0.72, and the probability of selecting a visitor who is located in room B is 0.24. How many visitors are located in room C?

(A) 4

(B) 9

(C) 39

(D) 108

Answer: B

286 Mark for Review

A model predicts that the population of a certain town was $7{,}000$ in 2007. The model also predicts that each year for the next 5 years, the population p increased by 5% of the previous year's population. Which equation best represents this model, where x is the number of years after 2007, for $x \leq 5$?

(A) $p = 0.95(7{,}000)^x$

(B) $p = 1.05(7{,}000)^x$

(C) $p = 7{,}000(0.95)^x$

(D) $p = 7{,}000(1.05)^x$

Answer: D

287 Mark for Review

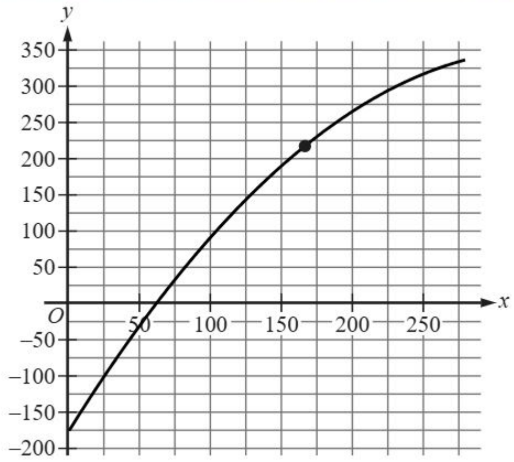
$$(x + 3)(x - 13) = 0$$

What is the positive solution to the given equation?

13

Answer: (grid-in: 13)

288 Mark for Review



The graph shows the estimated boiling point y, in degrees Celsius, of a straight-chain alkane with a molecular weight of x grams per mole, where $1 \leq x \leq 280$. Which statement is the best interpretation of the point $(167.15, 216.33)$?

Ⓐ A straight-chain alkane with a molecular weight of 216.33 grams per mole has an estimated boiling point of 167.15 degrees Celsius.

Ⓑ A straight-chain alkane with a molecular weight of 167.15 grams per mole has an estimated boiling point of 216.33 degrees Celsius.

Ⓒ The minimum estimated boiling point for straight-chain alkanes corresponds to an alkane with a molecular weight of 167.15 grams per mole and an estimated boiling point of 216.33 degrees Celsius.

Ⓓ The maximum estimated boiling point for straight-chain alkanes corresponds to an alkane with a molecular weight of 167.15 grams per mole and an estimated

Answer: B

289 Mark for Review

Line t in the xy-plane has a slope of $\frac{1}{8}$ and passes through the point $(40, -9)$. Which equation defines line t?

(A) $y = -14x + \frac{1}{8}$

(B) $y = \frac{x}{8} - 14$

(C) $y = \frac{x}{8} - 9$

(D) $y = 40x - 9$

Answer: B

290 Mark for Review

Each angle of $\triangle ABC$ is congruent to one of the angles of $\triangle QRS$. If $AB = 38$, $BC = 57$, $AC = 76$, and $RS = 228$, what is one possible value for the perimeter of $\triangle QRS$?

513

Answer: (grid-in: 513)

291 Mark for Review

The function g is defined by $g(x) = 4x + 8$, where $g(a) = 28$ and $g(4) = b$. What is the value of $a + b$?

(A) 144

(B) 32

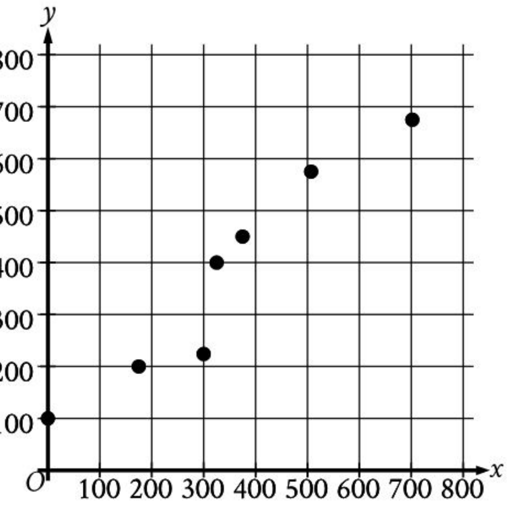
(C) 29

(D) 9

Answer: C

292 Mark for Review

The scatterplot shows the relationship between two variables, x and y.



Which of the following equations is the most appropriate linear model for the data shown?

(A) $y = 70.20 + 0.89x$

(B) $y = 0.89 + 70.20x$

(C) $y = 70.20x$

(D) $y = -0.89x$

Answer: A

293 Mark for Review

A right square prism has a height of 25 units. The volume of the prism is $2,025$ cubic units. What is the length, in units, of an edge of the base?

(A) 9

(B) 25

(C) 45

(D) 81

Answer: A

294 Mark for Review

$$-3x + 24px = 48$$

In the given equation, p is a constant. The equation has no solution. What is the value of p?

(A) 0

(B) $\frac{1}{8}$

(C) $\frac{2}{3}$

(D) 2

Answer: B

295 Mark for Review

If $x \geq -8$ represents all solutions to the inequality $ax - 28 \leq 12$, where a is a constant, what is the greatest possible value of a?

-5

Answer: (grid-in: -5)

296 Mark for Review

$$y = 3x^2 + 22$$
$$y = px + 19$$

In the given system of equations, p is a positive constant. The graphs of the equations in the given system intersect at exactly one point, (x, y), in the xy-plane. What is the value of x?

(A) 41

(B) 25

(C) 6

(D) 1

Answer: D

297 Mark for Review

$$y < -6x - 19$$
$$y > -4x - 11$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given system of inequalities?

(A)

x	y
-5	7
-6	23
-8	-3

(B)

x	y
-5	8
-6	15
-8	27

(C)

x	y
-5	10
-6	16

Answer: C

298 Mark for Review

The function f is defined by $f(x) = 41(0.26)^x$. For any positive integer n, the value of $f(n)$ is $p\%$ less than the value of $f(n-1)$. What is the value of p?

(A) 74

(B) 59

(C) 41

(D) 26

Answer: A

299 Mark for Review

A community organization surveyed a random sample of residents of the community to estimate the percentage of residents who visit their local library at least once per month. From the survey, the organization estimates that 12% of residents of the community visit their local library at least once per month, with an associated margin of error of 3.9%. The organization repeated the survey with a random sample of residents that was double the original sample size. Assuming the margins of error are calculated in the same way for both studies, which of the following is true about the margin of error associated with the estimate from the larger sample?

A. The margin of error is less than 3.9%.

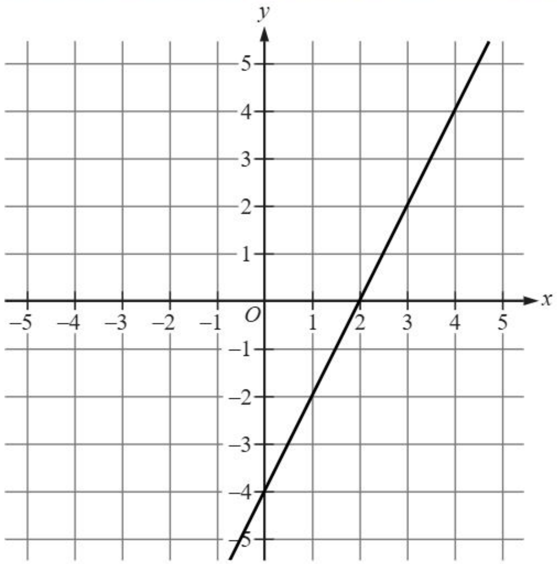
B. The margin of error is 7.8%.

C. The margin of error is greater than 12%.

D. The margin of error is 24%.

Answer: A

300 Mark for Review



The line shown can be represented by the equation $ax + by = 28$, where a and b are constants. Which of the following is true about a and b?

(A) $a < 0$ and $b < 0$

(B) $a < 0$ and $b > 0$

(C) $a > 0$ and $b < 0$

(D) $a > 0$ and $b > 0$

Answer: C

301 Mark for Review

$$2x = 7 + 8y$$
$$-2x = 4 + 8y$$

The solution to the given system of equations is (x, y). What is the value of $4x$?

3

Answer: (grid-in: 3)

302 Mark for Review

$$-5x^2 + bx - 45 = 0$$

In the given equation, b is a positive integer. The equation has no real solution. What is the largest possible value of b?

29

Answer: (grid-in: 29)

303 Mark for Review

The function f is a quadratic function. In the *xy*-plane, the graph of $y = f(x)$ has a vertex at $(1, 7)$ and passes through the points $(2, 40)$ and $(-1, 139)$. What is the value of $f(-2) - f(0)$?

Ⓐ 106

Ⓑ 172

Ⓒ 264

Ⓓ 304

Answer: C

304 Mark for Review

Each 3.0-ounce serving of cheddar cheese and each 1.2-ounce serving of tuna provides about 1 microgram of vitamin B12. If a total of 3.9 micrograms of vitamin B12 are consumed from eating x ounces of cheese and y ounces of tuna, which equation best represents this situation?

(A) $0.83x + 0.33y = 3.9$

(B) $1.2x + 3.0y = 3.9$

(C) $3.0x + 1.2y = 3.9$

(D) $0.33x + 0.83y = 3.9$

Answer: D

305 Mark for Review

The expression $\frac{kx-42}{2x^2-11x+15}$ is equivalent to $\frac{p}{x-3}-\frac{1}{2x-5}$, where k and p are constants. What is the value of p?

(A) -41

(B) $\frac{39}{5}$

(C) 9

(D) 17

Answer: C

306 Mark for Review

A zoologist observed the nests of wood ducks in an area. Each year, the zoologist recorded the number of eggs in each of the first 55 nests that were observed and created a frequency table for the data set. Which of the following frequency tables represents the data set with the smallest standard deviation?

Ⓐ

Number of eggs	Frequency
11	12
12	11
13	9
14	11
15	12

Ⓑ

Number of eggs	Frequency
11	11
12	11
13	11
14	11

Answer: D

307 Mark for Review

The area of a rectangular region is increasing at a rate of 330 square feet per hour. Which of the following is closest to this rate in square meters per minute? (Use $1 \text{ meter} = 3.28 \text{ feet}$.)

(A) 0.51

(B) 1.68

(C) 18.04

(D) 30.67

Answer: A

308 Mark for Review

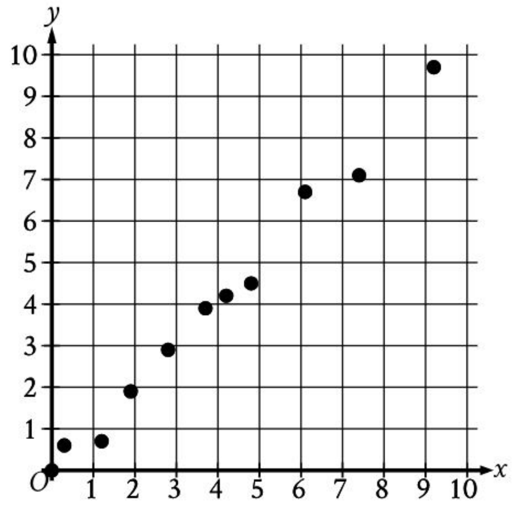
A researcher investigated two species of mites: a predator and its prey. At the start of a week, there was an equal number of the two species. At the end of the week, the number of prey had increased by 2100% of the number of prey at the start of the week, and the number of predators had increased by 450% of the number of predators at the start of the week. The number of predators at the end of the week was $p\%$ less than the number of prey at the end of the week. What is the value of p?

75

Answer: (grid-in: 75)

309 Mark for Review

The scatterplot shows the relationship between two variables, x and y.



Which of the following equations is the most appropriate linear model for the data shown?

Ⓐ $y = 0.07 - 1.03x$

Ⓑ $y = 1.03 - 0.07x$

Ⓒ $y = -0.07 + 1.03x$

Ⓓ $y = -1.03 + 0.07x$

Answer: C

310 Mark for Review

A model predicts that the population of a certain town was $20{,}000$ in 2002. The model also predicts that each year for the next 5 years, the population p increased by 6% of the previous year's population. Which equation best represents this model, where x is the number of years after 2002, for $x \leq 5$?

(A) $p = 0.94(20{,}000)^x$

(B) $p = 1.06(20{,}000)^x$

(C) $p = 20{,}000(0.94)^x$

(D) $p = 20{,}000(1.06)^x$

Answer: D

311 Mark for Review

$$4x + 4y = 6$$
$$6x - 4y = 36$$

In the *xy*-plane, the graphs of the given equations intersect at (x, y). What is the value of $x + y$?

3/2

Answer: (grid-in: 3/2)

312 Mark for Review

For the linear function f, the graph of $y = f(x)$ in the xy-plane has a slope of 0 and passes through the point $(0, 14)$. Which equation defines f?

(A) $f(x) = x + 14$

(B) $f(x) = 14x$

(C) $f(x) = 0$

(D) $f(x) = 14$

Answer: D

313 Mark for Review

Triangle ABC is similar to triangle XYZ such that A, B, and C correspond to X, Y, and Z, respectively. The length of each side of triangle ABC is n times the length of its corresponding side in triangle XYZ, where n is an integer greater than 1. The measure of angle C is 59 degrees, and $YZ = 74$. Which of the following must be true?

A) $BC = \frac{74}{n}$

B) $BC = 74n$

C) The measure of angle Z is $\frac{59}{n}$ degrees.

D) The measure of angle Z is $59n$ degrees.

Answer: B

314 Mark for Review

$$5 - 4y = x$$
$$7 + 4y = x$$

The solution to the given system of equations is (x, y). What is the value of x?

6

Answer: (grid-in: 6)

315 Mark for Review

The function g is defined by $g(x) = 4x + 6$, where $g(a) = 26$ and $g(4) = b$. What is the value of $a + b$?

(A) 132

(B) 30

(C) 27

(D) 9

Answer: C

316 Mark for Review

The function g is defined by $g(x) = -2x(x+3)(x-k)^2 + r$, where k and r are integer constants. In the xy-plane, the graph of $y = g(x)$ passes through the point $(7, 13)$, and $g(0) = 13$. What is the value of $r + k$?

Ⓐ -7

Ⓑ 6

Ⓒ 13

Ⓓ 20

Answer: D

317 Mark for Review

A chef made 9 cups of sauce. The chef then filled x small jars and y large jars with all the sauce he made. The equation $4x + 5y = 9$ represents this situation. Which is the best interpretation of $5y$ in this context?

(A) The number of large jars the chef filled

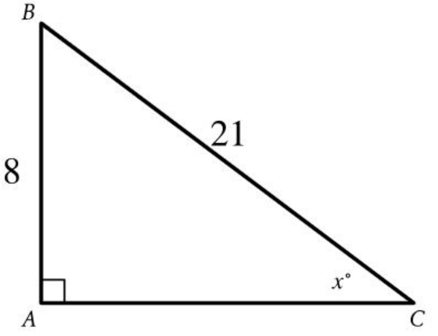
(B) The number of small jars the chef filled

(C) The total number of cups of sauce in the large jars

(D) The total number of cups of sauce in the small jars

Answer: C

318 Mark for Review



Note: Figure not drawn to scale.

The expression $k \cdot \cos x^{\circ}$ represents the area of triangle ABC shown. What is the value of k?

84

Answer: (grid-in: 84)

319 Mark for Review

$$f(x) = (2x - 13)(2x + 5)$$

What is the minimum value of the given function?

A. -81

B. $-\frac{5}{2}$

C. 2

D. 8

Answer: A

320 Mark for Review

$$\frac{1}{34} = \frac{1}{R_1} + \frac{1}{R_2}$$

The given equation relates the resistances, R_1 and R_2, in ohms, of two resistors arranged in parallel in a circuit with a total resistance of 34 ohms, where R_1 and R_2 are positive. Which equation correctly expresses R_1 in terms of R_2?

A) $R_1 = 34 + R_2$

B) $R_1 = 34 - R_2$

C) $R_1 = \frac{34R_2}{R_2+34}$

D) $R_1 = \frac{34R_2}{R_2-34}$

Answer: D

321 Mark for Review

$$y \geq x + 15$$
$$y \leq -5x - 15$$

Which point (x, y) is a solution to the given system of inequalities in the xy-plane?

(A) $(-4, 8)$

(B) $(-4, 11)$

(C) $(-6, 8)$

(D) $(-6, 11)$

Answer: D

322 Mark for Review

x	y
$-2s$	21
$-s$	17
s	9

The table shows three values of x and their corresponding values of y, where s is a constant. There is a linear relationship between x and y. Which of the following equations represents this relationship?

(A) $sx + 4y = 13s$

(B) $4x + sy = 13s$

(C) $4x + sy = 13$

(D) $sx + 4y = 13$

Answer: B

323 Mark for Review

Which expression is **NOT** a factor of $1{,}120x^4 - 43{,}750$?

(A) $4x^2 + 25$

(B) $2x^2 - 5$

(C) $2x + 5$

(D) 70

Answer: B

324 Mark for Review

The percent increase in mass of a certain red kangaroo from 150 days old to 230 days old was 422%. If this red kangaroo's mass was k grams at 150 days old, which expression represents its mass, in grams, at 230 days old?

(A) $0.05k$

(B) $1.05k$

(C) $4.22k$

(D) $5.22k$

Answer: D

325 Mark for Review

Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is 46 square centimeters (cm^2), and the surface area of right rectangular prism Y is $736\ \text{cm}^2$. The volume of right rectangular prism Y is $1{,}280$ cubic centimeters (cm^3). What is the sum of the volumes, in cm^3, of right rectangular prism X and right rectangular prism Y?

1300

Answer: (grid-in: 1300)

326 Mark for Review

$$(x-k)^2=(k-4a)(x-k)$$

In the given equation, a and k are constants, where $k>4a$. The sum of the solutions to the equation is $3k+37$. What is the value of a?

-37/4

Answer: (grid-in: -37/4)

327 Mark for Review

An artist made 12 pieces of pottery using 30 pounds of clay. The pottery included only bowls and vases. Each bowl used 2 pounds of clay and each vase used 4 pounds of clay. How many vases did the artist make?

A) 3

B) 5

C) 7

D) 9

Answer: A

328 Mark for Review

A bookstore is giving away free books every day. The equation $f(x) = 160 - 10x$ gives the number of free books, $f(x)$, that remain to be given away after x days. Which of the following is the best interpretation of $f(16) = 0$ in this context?

(A) The bookstore has 16 free books to give away.

(B) The bookstore will give away 16 free books in 10 days.

(C) It will take 16 days to give away all the free books.

(D) It will take 160 days to give away 16 free books.

Answer: C

329 Mark for Review

The table gives the distribution of flavor and type of topping for customer orders at an ice cream shop.

Flavor of ice cream	Type of topping	
	Sprinkles	No Sprinkles
Chocolate	40	50
Vanilla	30	10
Twist	70	30

If a customer order is selected at random, what is the probability of selecting an order with sprinkles, given the flavor of ice cream is vanilla?

A) 0.13

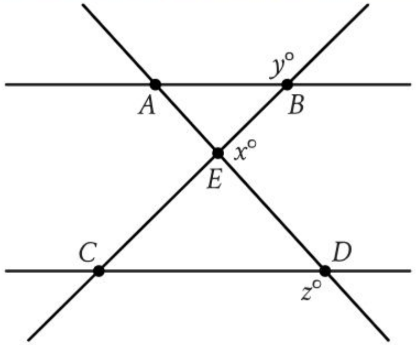
B) 0.21

C) 0.43

D) 0.75

Answer: D

330 Mark for Review



Note: Figure not drawn to scale.

In the figure, line AB is parallel to line CD, and line AD intersects line BC at point E. If $y = 114$ and $z = 119$, what is the value of x?

(A) 53

(B) 61

(C) 127

(D) 143

Answer: C

331 Mark for Review

Which expression is equivalent to $\frac{1}{2y^9}$, where $y > 1$?

Ⓐ $2y^{-9}$

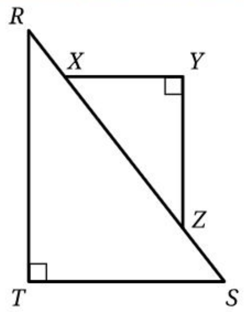
Ⓑ $(2y)^{-9}$

Ⓒ $\frac{2}{y^{-9}}$

Ⓓ $\frac{y^{-9}}{2}$

Answer: D

332 Mark for Review



Note: Figure not drawn to scale.

In triangles RST and XYZ shown, $\overline{XY}$ is parallel to $\overline{TS}$ and $\tan R = \frac{96}{247}$. What is the value of $\sin X$ in triangle XYZ?

(A) $\frac{96}{343}$

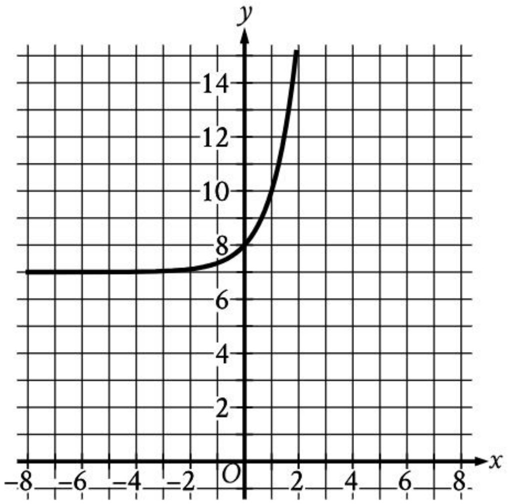
(B) $\frac{96}{265}$

(C) $\frac{247}{265}$

(D) $\frac{247}{96}$

Answer: C

333 Mark for Review



What is an equation of the graph shown?

Ⓐ $y = 3^x + 7$

Ⓑ $y = 3^{-x} + 7$

Ⓒ $y = 3^x + 8$

Ⓓ $y = 3^{-x} + 8$

Answer: A

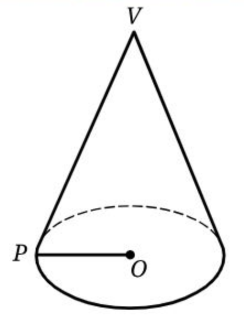
334 Mark for Review

A company developed a plan to set the selling price of a product. The company determined that for a selling price of $\$90.00$, zero products would be sold. For each $\$1.50$ decrease in the selling price, the number of products sold would increase by one. For a revenue of exactly $\$1,336.50$, which of the following could be the number of products sold? $(\text{revenue} = \text{price} \times \text{number of products sold})$

- (A) 27
- (B) 30
- (C) 831
- (D) 1,350

Answer: A

335 Mark for Review



Note: Figure not drawn to scale.

For the right circular cone shown, $\overline{OP}$ is a radius of the base and V is the vertex. The cone's lateral surface area, in square meters, is $rL\pi$, where r is the length, in meters, of $\overline{OP}$ and L is the length, in meters, of $\overline{VP}$. The cone has a base area of 16π square meters and a lateral surface area of 51π square meters. What is the height of the cone, to the nearest meter?

12

Answer: (grid-in: 12)

336 Mark for Review

The function f is a quadratic function. In the *xy*-plane, the graph of $y = f(x)$ has a vertex at $(1, 4)$ and passes through the points $(2, 52)$ and $(-1, 196)$. What is the value of $f(-2) - f(0)$?

Ⓐ 148

Ⓑ 244

Ⓒ 384

Ⓓ 436

Answer: C

337 Mark for Review

$$f(x) = 23(1.20)^{\frac{x}{4}}$$

For the given function f, the value of $f(x)$ increases by $p\%$ for every increase of x by 8. What is the value of p?

44

Answer: (grid-in: 44)

338 Mark for Review

The area of a rectangular region is increasing at a rate of 260 square feet per hour. Which of the following is closest to this rate in square meters per minute? (Use $1 \text{ meter} = 3.28 \text{ feet.}$)

(A) 0.40

(B) 1.32

(C) 14.21

(D) 24.17

Answer: A

339 Mark for Review

The product of a number and $-\frac{2}{19}$ is 1. What is the number?

Ⓐ $\frac{19}{2}$

Ⓑ $\frac{2}{19}$

Ⓒ $-\frac{2}{19}$

Ⓓ $-\frac{19}{2}$

Answer: D

340 Mark for Review

The height of a right circular cylinder is 21 inches, and the circumference of its base is 210 inches. Which expression represents the total surface area, in square inches, of the cylinder?

(A) $(21)(210) + 2\pi\left(\frac{105}{\pi}\right)^2$

(B) $(21)(210) + \pi\left(\frac{105}{\pi}\right)^2$

(C) $\pi\left(\frac{105}{\pi}\right)^2(21)$

(D) $(21)(210)$

Answer: A

341 Mark for Review

The function f is defined by $f(x) = |x - 6x|$. What value of a satisfies $f(1) - f(a) = -15$?

Ⓐ -16

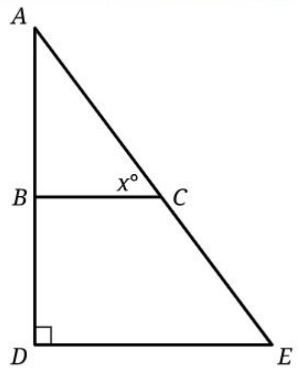
Ⓑ 3

Ⓒ 4

Ⓓ 75

Answer: C

342 Mark for Review



Note: Figure not drawn to scale.

In the figure, $\overline{BC}$ is parallel to $\overline{DE}$. If the length of $\overline{DE}$ is 162 and the length of $\overline{AE}$ is 270, what is the value of $\tan x^\circ$?

(A) $\frac{270}{162}$

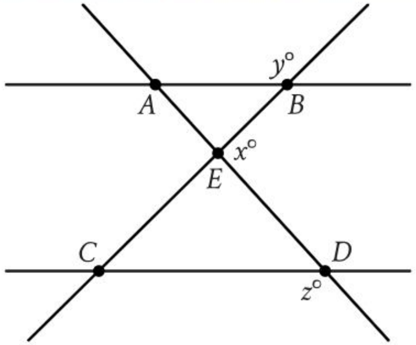
(B) $\frac{216}{162}$

(C) $\frac{162}{270}$

(D) $\frac{162}{216}$

Answer: B

343 Mark for Review



Note: Figure not drawn to scale.

In the figure, line AB is parallel to line CD, and line AD intersects line BC at point E. If $y = 111$ and $z = 118$, what is the value of x?

Ⓐ 49

Ⓑ 62

Ⓒ 131

Ⓓ 139

Answer: C

344 Mark for Review

$$y = 4x^2 + 43$$
$$y = px + 39$$

In the given system of equations, p is a positive constant. The graphs of the equations in the given system intersect at exactly one point, (x, y), in the xy-plane. What is the value of x?

Ⓐ 82

Ⓑ 47

Ⓒ 8

Ⓓ 1

Answer: D

345 Mark for Review

A car dealership has only sedans, SUVs, and minivans for sale. On Monday, 20% of the vehicles for sale were sedans and 50% were SUVs. If there were 36 sedans for sale at the dealership on Monday, how many minivans were for sale?

54

Answer: (grid-in: 54)

346 Mark for Review

An exponential function f gives the estimated amount of an X-ray beam's initial intensity remaining after passing through a w-centimeter-thick window made of beryllium. The function estimates that after passing through a 1.8-centimeter-thick window, the amount of the X-ray beam's initial intensity remaining is 0.65. Which equation could define f?

(A) $f(w) = 0.65(0.79)^{w-1.8}$

(B) $f(w) = 0.65(0.79)^{\frac{w}{1.8}}$

(C) $f(w) = 0.65(1.8)^{w}$

(D) $f(w) = 1.8(0.65)^{w}$

Answer: A

347 Mark for Review

The function f is a quadratic function. In the *xy*-plane, the graph of $y = f(x)$ has a vertex at $(1, 5)$ and passes through the points $(2, 52)$ and $(-1, 193)$. What is the value of $f(-2) - f(0)$?

- (A) 146
- (B) 240
- (C) 376
- (D) 428

Answer: C

348 Mark for Review

Each 3.0-ounce serving of cheddar cheese and each 1.2-ounce serving of tuna provides about 1 microgram of vitamin B12. If a total of 2.4 micrograms of vitamin B12 are consumed from eating x ounces of cheese and y ounces of tuna, which equation best represents this situation?

(A) $0.83x + 0.33y = 2.4$

(B) $3.0x + 1.2y = 2.4$

(C) $1.2x + 3.0y = 2.4$

(D) $0.33x + 0.83y = 2.4$

Answer: D

349 Mark for Review

The area of a rectangular region is increasing at a rate of 230 square feet per hour. Which of the following is closest to this rate in square meters per minute? (Use 1 meter $= 3.28$ feet.)

Ⓐ 0.36

Ⓑ 1.17

Ⓒ 12.57

Ⓓ 21.38

Answer: A